

2 Imagined worlds

The politics of future-making in the twenty-first century

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The rise of science and technology as major forces in world-making upset three widely held assumptions about the foundations of politics: its materiality, its presentism, and its localism. Politics was traditionally thought to be a struggle over the conditions of everyday life in the present (Thompson 2010), a zero-sum game in which benefits to some arrive inevitably at cost to others. Conventional politics plays out, on this view, over the allocation and distribution of a bounded set of goods in a temporally and spatially contained world, among groups competing for their proverbial ‘share of the pie’. That pie is a package of benefits and burdens whose dimensions at any given moment are imagined as expressly known and calculable, and hence divisible among competing interests in ways that a polity can both see and understand. A simple example of this politics of instant gratification is the tax cut. One can observe almost instantly in a given year’s tax returns who won and who lost vis-à-vis the prior year, and whether promises were kept or not.¹ Outcomes of such politics, the testable politics of the here and now, are conceived as immediate, tangible, and available for popular evaluation and judgement. They most easily achieve what the political scientist Fritz Scharpf termed ‘output legitimacy’, or validation by results (Scharpf 1997).

Another adage holds that ‘all politics is local’. Political struggle, on this view, occurs not only in real time for shares of limited resources but also within a well-demarcated social environment: whether a relatively small one like a family, a school district or a municipality, or a much larger one like a nation state or, at the outer limit, the entire global community and even future generations. Issues closer to home typically offer both what Scharpf termed ‘input legitimacy’, an accepted process for aggregating preferences, and ‘output legitimacy’ in the form of visible results. Thus, mayors in northern American cities are routinely held accountable for failures of snow removal. One analyst of municipal politics remarked, only partly tongue in cheek: ‘But woe betide the big-city mayor who fumbles the fundamental test of municipal governance: snow ploughing. For that, there will be no forgiveness’ (Dudley 2017). On a less mundane scale, in American politics, Hubert Humphrey may have lost the 1968 presidential election to Richard Nixon because of televised riots and a perceived breakdown of law and order in Chicago. The failure of

George W. Bush's administration to cope with the aftermath of Hurricane Katrina in New Orleans may well have cost the Republican Party more votes in 2008 than the ill-fated Iraq war or that election year's fateful financial meltdown.

The space bounded by presentism and localism is where ideology historically took root, finding systematic grounds to favour one set of interests over another – whether for reasons of class identification, or a promised ability to make economies grow, or to defend transcendental values such as equality and social justice. Some struggles came to be seen as foundational enough to define politics, such as labour versus capital, privatization versus state control, independence versus colonial domination, and rights of disadvantaged groups versus majority entitlements. Sharp battle lines along these axes, dividing a canonical progressive left from an equally canonical reactionary right, presupposed that the salient factors determining political choice are known, or in principle knowable to all relevant actors: such as the nature of one's identity, economic position, and class interests.

The quintessential locus of democratic politics, on the classical account, is the polling place or voting booth. This is where the citizens of a *polis*, those formally entitled to participate in the political process, record their preferences, elect representatives, and authorize actions that the majority desires. Elections are occasions where having a political voice counts and political belonging can be performed. Hence, too, the recurrent great movements of emancipation, especially over the long twentieth century, to bring historically excluded groups into the ranks of voting citizens, as acknowledged members of a political community. Persistent denial of such representation has lit the fuse for revolutionary protests the world over, whether in the United States in 1775, 1861, or 1920, France in 1789, India in 1930, or Hong Kong in 2019.²

In contemporary politics, the street and the polling place serve as complementary vessels for channelling the same basic demand for voice and recognition. Both accommodate great outpourings of popular sentiment to express how people wish to be governed with respect to their matters of concern. But whereas the voting booth is open to every recognized member of a democratic polity, the street attracts those upstarts and renegades who feel they do not yet have a voice in their community's official forms of politics. One thrives on routinized process and an established past; the other on imagination and promises, and a future still in the making. One represents the taming of democracy, the other its unchained force (Hardt and Negri 2000).

Now, as the late twentieth century yields to the twenty-first, rents in our traditional understanding of politics have become apparent. First, political theory has tended to ignore a fundamental question about what makes people feel (or not feel) part of a given polity, that is, the nature of subjectivity and belonging that are constitutive of political identities and movements. What is it about political membership – not merely an imposed, top-down nationalism (Anderson 2006[1983]) – that means something to people, enough to make them go out and vote or to campaign for representation? Voting behaviour is especially puzzling.

It is only in relatively authoritarian nations, or where compulsory voting is enforced, that a high percentage of the electorate turns to the ballot box. Elsewhere, apathy is more often the rule, and local elections where issues are presumably closest to home often draw the lowest levels of participation. Voting, in short, seems a high-level political desideratum exactly where one does not have it as a right, but at best an ambiguous good in societies where the right is freely given.³ This suggests that, even prior to the politics of exercising voice (Hirschman 1970), there must be a politics of identity-making, coupled with a sense of purpose and engagement – a judgement that there is something important to come together and vote *for* – that has received too little attention from analysts of the politics of the present.

Further, the issues and concerns that mobilize people in contemporary societies seem often to have little to do with grand questions of right or left ideology. Since the political shocks of 2016 – Britain’s withdrawal from the European Union, the election of US president Donald J. Trump, and the rise of disaffected populist parties throughout Europe and beyond – commentators have wondered why people vote against their long-term class or economic interests. Some have blamed the rise of a post-truth culture and a loss of trust in expertise for the surge of populism (d’Ancona 2017). Others point to the disconcerting gap between rhetoric and realization. Disappointed expectations of immediate gratification and local order seem to mobilize political responses more effectively than abstract claims about liberty or human rights or even progress. If there is one dominant ideology that cuts across large swaths of the globe in this century, it is a return to nativism, a rebellion against all of the mid-century institutions built to ensure global peace, prosperity, and well-being. The practices of early twenty-first century politics seem in this respect to belie the confident predictions of both economic expertise and grand theories such as world systems analysis (Wallerstein 2004). If transnational collectives are forming today, it is not around ideas of economic growth or the global common good; if anything, such movements may be driven more by fear than by optimism.

Perhaps most significantly, the zero-sum model of politics and redistribution takes little account of the role of science and technology in creating new horizons of benefit and risk. Scientific discoveries and their applications have opened up the future as a space of political struggle in countless ways. From the lure of increasing automation, powered by immense computing capability, to the eradication of unwanted life forms, genetic disease, and even aging, science and technology offer promises of futures that seem worth fighting for, or possibly against, depending on actors’ visions of what makes societies good and desirable (Jasanoff and Kim 2015). Technology, moreover, is a harbinger of extreme and catastrophic visions as well as progressive ones. An era marked by threats of nuclear and climate annihilation, and the imminent takeover of the human workforce by machines, has altered the stakes of global politics and created new political alliances, bypassing or even excluding the state actors who previously dominated the international stage.

Indeed, contestation over possible technoscientifically driven futures turns several key assumptions of traditional interest-group politics on their head. In the futuristic world, the pie of benefits is not necessarily limited in size, because goods such as a cure for cancer will theoretically be available to all humanity, just as threats of extreme disasters afflict us all (Beck 1992[1986]). Further, matters in deep contestation, such as the impacts of climate change or increasing automation, are not only of concern to present generations; nor, in the digital era, are sites of struggle necessarily bounded by the territorial borders of particular geopolitical entities, such as a city, region, or nation. More importantly, the speculative character of prediction breaks apart the relatively simple synergy between knowing and acting that powered classical political mobilization. The consequences and questions for politics are profound.

As if in a Jenga game, the familiar supports for shaping and theorizing politics have been pulled out one by one in the politics of future-making, leaving analysts to question whether long-established political forms, even democracy itself, are any longer viable in the spatial and temporal orders of prediction. In the dim, partially discernible spaces of what is to come, what takes the place of mundane, visible, real-world outputs, such as taxes or potholes or snow removal? If the benefits and burdens to be allocated through struggle lie in the realm of speculation, then how do people gain enough confidence to act on such predictions? And who in any case are 'the people'? Without a recognizable *polis* to respond to impending global crises, how can there be a well-defined polity to engage meaningfully in political action?

These unsettling aspects of predictive politics are the subject of this volume. The book as a whole seeks to understand how science and politics interact to create spaces of political involvement that lack the familiar markers of certainty about the purposes and means of struggle, and yet are replete with issues of entitlement and allocation that are the very stuff of politics. These are spaces in which imagination becomes a potent resource, since persuading people to opt for any vision of a future world, *the things to come* as the introduction puts it, requires a leap into a fictive, unrealized landscape of dreams rather than one of tangible reality. At the same time, those visions must have enough solidity to win people's allegiance, to enable action, and secure buy-in. In this chapter, using analytic resources from the field of science and technology studies (STS), I look at three sites in which the dynamics of predictive politics is playing out: making the objects of governance, making instruments of intervention, and making political subjects and polities. In each case, science, expertise, and material technologies function as agents of social and political mobilization, alongside and intertwined with traditional political institutions such as legislatures and regulatory agencies. Overall, I argue that the rise of prediction as a mode of governance has troubled existing practices of constitutional government, thrust new questions of trust and accountability into national and international debate, and set in train movements in political and social thought that amount, in effect, to a reconstitutionalizing of contemporary politics.

Socio-technical imaginaries: theorizing the future

Economic activity has long been modelled as the process of production, based on the Marxian factors of land, labour, and capital. Technological innovation and improvements can alter the relationships among those classic input variables, producing not only new constellations of economic power (or dispossession) but also changes in the social order as power moves from the haves to the have-nots, and vice versa. Thus, the transition from a coal- to an oil-based economy in the twentieth century brought large shifts in the socio-economic status of nations and regions, lifting once-poor countries like Norway and the Gulf States to the richest of rich, but also as Timothy Mitchell has argued, disrupting the place-based foundations of democratic politics (Mitchell 2011). With climate change putting fossil-fuelled futures in jeopardy, wind and sunshine along with the capacity to harness them emerged as new forms of wealth and power, supplementing decades-old efforts to enlist atoms for peace with nuclear technologies. All these developments still rest on the exploitation of land, labour, and capital, although the forms and modes of capitalization have shifted in politically dramatic ways.

That politics too can be viewed as a site of production, relying on a range of input variables, is less commonly discussed than are models of economic growth; and yet in the era of prediction the development of science and technology goes hand in hand with the capacity to produce new political imaginations. The two forms of productivity synergize, and the resulting dynamic is what STS scholars term 'co-production'. It is the play of giving shape simultaneously to a material world through technoscientific means and to political ideas of how societies should respond to the resulting order of things. In my introductory essay in *States of Knowledge*, I defined the term as follows: 'Co-production is shorthand for the proposition that the ways in which we know and represent the world (both nature and society) are inseparable from the ways in which we chose to live in it' (Jasanoff 2004: 2). Co-production can be observed at all possible scales of governance and in many processes of social construction, most especially in the making of representations, discourses, identities, and institutions – in short in the things to be governed, in the instruments that do the work of governing, and in the politics with stakes in the legitimacy of government.

Seeing technologies as instruments of political as well as material production complicates a familiar argument from the early days of science and technology studies, namely, in the words of political scientist Langdon Winner, that 'artifacts have politics' (Winner 1980). Winner suggested that the very design of technological systems, such as their relative opacity or transparency, aligns with particular forms of politics. A highly black-boxed and inscrutable technological system, such as nuclear power generation, lacks transparency, Winner argued, and thus deprives citizens of meaningful opportunities to take part in managing it. The same could be said of something as simple as a highway design that lets some vehicles freely get to some locations but inhibits other forms and routes of travel.⁴ Technology's unchecked power over our lives, exercised through often

invisible design choices, is inconsistent with democracy's first principle: That members of a polity must have opportunities to penetrate the workings of power, demand explanations, and say yea or nay to the choices made by governing political institutions. Anything else would be plainly anti-democratic.

Winner's way of thinking about technology provided an easy conceptual bridge between two realms that political science had traditionally held apart, the technological (or in more contemporary language, the technoscientific) and the political. His succinct formulation insisted that technological design serves a political purpose because it enables some ways of doing things and disables others, and because those choices are not always plain to the *demos*. Still less are technological choices deliberated or granted popular assent. Yet the simplicity of Winner's analysis was also a problem, in that attributing 'politics' to 'artifacts' represented both worlds as more static and invariant than they are in practice. Politics, after all, comes in many flavours, as do technological systems, and both evolve and change through their interaction. Merely pointing out that the two are linked says nothing about how the one affects the other, nor why the same technological system elicits radically different forms of politics and governance in different cultural settings (Jasanoff and Kim 2009). To explain the diverse ways in which technological trajectories develop around the world, one needs a more nuanced theory, one grounded in a deep understanding of co-production and sensitive to the distinctive attributes of the politics of future-making. The concept of socio-technical imaginaries (STIs) offers such a framework.

As defined in *Dreamscapes of Modernity* (Jasanoff and Kim 2015: 4), STIs are 'collectively held, institutionally stabilized, and publicly performed visions of desirable futures, animated by shared understandings of forms of social life and social order attainable through, and supportive of, advances in science and technology'. This definition brings under one roof Benedict Anderson's famed idea of nationhood as a kind of 'imagined community' (Anderson 2006[1983]: 6) and the efforts of STS scholars such as me to show that collective political imaginations today cannot be separated from the futures promised by science and technology. It follows, too, that the contemporary orchestrators of collective visions are as much leaders of science and industry as minions of the state. Anderson identified the state and its captive print media as the primary agents of creating national fellow-feeling.⁵ Adding science and technology to the mix decentres the state, as well as the media, although alliances between states and science remain, as ever, powerfully constitutive (see, for example, Edgerton 2011). Science, however, brings its own independent ideas of progress and salvation to the building of collective consciousness, and the resulting forms of self-understanding and self-identification are not necessarily congruent with the nationalisms of earlier centuries. Importantly, then, unlike prior work on imagined national communities, the STI framework attends to just the sorts of transformations that make the politics of the twenty-first century so different from the politics that went before, detaching it from tangible materiality, temporal immediacy, and spatial boundedness. These shifts have left their mark on the

definition of objects and instruments of governance, the composition of polities, and of course the age-old question of who rules.

Governable objects in the era of prediction

Possibly no single object now known to humankind has demonstrated quite such capacity to reshape politics as the global climate. In relatively short order, humanity's view of its appointed fate shifted from thoughts of nuclear annihilation (Weart 1988) to the spectre of civilization ending through a drastic rise in the Earth's mean surface temperature. Attempts to come to grips with the facts of climate change altered how we think about the weather forecast and our own consumption habits, has given rise to new institutions and discourses of management and control, empowered new cadres of experts, created markets, and fostered improbable alliances between and across industrial sectors and nations. Yet, for all its power to mobilize human action, the global climate remains a pure abstraction. Built out of decades, in some cases centuries, of observation of changes in the Earth's weather, soils, water, and vegetation, it is documented in countless data points and combined into a pervasive whole (Edwards 2010); yet it cannot be grasped in the hand like a loaf of bread, a tank of fuel, or a piece of paper bearing its monetary worth on its face. Nor can climate change be felt on the skin like the day's weather. In its immateriality, its placelessness, and its indeterminate impacts, the global climate is typical of the governable objects of future politics. But it is not the only such construct.

Long before climate rose to the forefront of global policy consciousness, there was the concept of risk. Threats to human well-being come in varying degrees of scope and significance: from *de minimis* or low-consequence ills, like a sprain from running or a fall from a ladder, to catastrophic and high-consequence events, such as a nuclear accident. The resulting harms can be acute or chronic, reversible or irreversible. All are subsumed under the concept of risk. Technically defined as the probability of harm times the magnitude of harm, risk evolved at one level as a mathematical concept, but for policy-makers it also spawned a discourse of legitimation, a way of rationalizing the exercise of power directed toward future ends. A landmark report of the US National Academy of Sciences recommended in 1983 that risk should be governed through a two-step process: assessment and management (NRC 1983). Risk assessment should be conducted largely on the basis of sound science, to determine the seriousness of the projected harm, whereas risk management should incorporate social values and weigh risks against benefits to arrive at appropriate policy responses (NRC 1983).

To the extent that risks befall members of societies, risk assessment belongs to the social sciences, and increasingly these are sciences of prediction. The rise of the quantitative social sciences from the mid-nineteenth century (Hacking 1990; Porter 1986) on through this century's developments in computer science, modelling and latterly data science made society's dynamics ever more tractable at temporal removes and scales of aggregation that pre-moderns could

not have imagined. What societies seek to govern, and how they govern it, were transformed. The aims and objects of governance morphed in size and scope, as well as in their temporal situation, from small and near-term problems – those potholes and snowy roads – to distant, geographically dispersed, and increasingly faraway futures.

To be sure, the ability to imagine problems beyond the immediate and the tangible was itself a form of co-production that evolved over centuries. From the beginnings of the Enlightenment, science has populated the human imagination with invisible objects that can nevertheless be made tractable through technological means, from the vacuum to the germ, the atom, and the gene. Edward Jenner and Louis Pasteur found ways to immunize people and animals against diseases carried by invisible infectious agents, and Pasteur developed a technique that rids milk of most of its deadly pathogens. Physicists probed the atom's internal structure, although it took the bomb and the nuclear reactor to make the atom's energy comprehensible. Penicillin and other antibiotics cured illness and prevented epidemics, while DDT and related chemicals tackled the animal vectors carrying lethal parasites. Faced with mounting evidence that unseen does not have to mean unmanageable, laws that once held people liable only for deterministic chains of causation⁶ gradually enlarged the scope of responsibility to include what experts could reasonably have foreseen. In one notorious episode, an Italian court in L'Aquila convicted half a dozen seismologists and public officials for failing to give due warning of a deadly earthquake in 2009. Critics of the (eventually overturned) judgement complained that it was based on a fundamental misconception of how much certainty science could provide, but many believed that experts should have told people that a catastrophic threat could not be ruled out. Citizens, they argued, could then have taken reasonable precautions and survived.

Such shifts in problem framing, from imminent to futuristic, entailed corresponding changes in people's expectations of how governing authorities should couch their arguments and justify their actions. Mathematical reasoning became pervasive and was increasingly identified with objective judgement (Porter 1995; Deringer 2018). At the same time, this dispassionate way of constructing dependable social knowledge (Camic *et al.* 2011) demoted people's subjective experiences of the world, creating tensions between lives lived in the present and lives not yet in being, governed by sophisticated mathematical predictions reaching into remote futures. Not everything that the predictive sciences rendered as equivalent for risk management were felt as commensurable by people on the receiving end of power and policy. How does one weigh the death of one's child from a rare adverse drug reaction against public health benefits for society as a whole (Calabresi and Bobbitt 1978), or the sacrifice of an indigenous nation's sacred lands against the benefits of cheap electricity for people to whom that land has no meaning (Espeland 1998)? Monetization added its own layers of uncertainty and conflict, as planners came to realize that cash in the hand now is worth more to most people than cash 10 or 20 years hence. Economists developed practices of discounting to decide what value to place in

the present on future expectations, thereby bringing within the scope of governance matters that had previously lodged outside the bounds of imagined human control (Beckert and Bronk 2018).

What, then, are the implications for politics, as immediacy recedes and governments engage more in remote threat management, possibly tilting at windmills, than in the delivery of tangible welfare benefits? One could point to the rise of sceptical populisms and newly reimagined nationalisms as a partial worldwide reaction to the detached futurism of the predictive natural and social sciences. The grand climate debates are the most prominent site where the battle lines are clearly drawn, with scientists, liberals, and rich-country youth joining hands on one side and reactionary politicians and their allies in the fossil fuel industry on the other. Each is invested in an existential future whose loss seems unthinkable, but the futures themselves are radically incompatible articulations of different worlds that combine the material and the moral substrates of life in ways that do not mesh. Sweden's Greta Thunberg spoke as if inspiring a latter day Children's Crusade when she stood before the United Nations General Assembly in 2019 and exhorted her elders to give back to her generation a future that seemed to be slipping away: 'I should be back in school on the other side of the ocean. Yet you all come to us young people for hope. How dare you. You have stolen my dreams and my childhood with your empty words.'⁷ Her example energized marches around the world and weekly convocations of young people convening under the banner of 'Fridays for Future', a future so seemingly cataclysmic as to rob children of their expected agency. Even education, the classic gateway to flexible futures, seemed useless to these crusading youngsters unless the climate crisis, as science predicts it, is brought under rapid control.

Yet, in the United States, the biggest source of greenhouse gas emissions, Republicans angrily dismissed Thunberg's message, while Donald Trump continued his tirades on behalf of an alternate reality, one in which climate change is a conspiracy and coal remains a clean and beautiful source of power. On this basis, Trump fought for industries that would have to shut down if Thunberg's exhortations were enacted into policy, leaving their employees, among the president's most reliable supporters, without a working future. In mining country, the prospect of closing the mines leaves already depressed economic zones without much promise of life beyond. Alternative pathways would require massive investments, even thinking anew the landscape itself. In the fracking, mining, and refining heartlands of the United States, no-one has yet conceived the sort of vast, yet incremental, geoengineering projects that Germany undertook to convert the old strip mines of former East Germany into new lake districts for well-to-do vacationers and leisure-seekers from the West (Gross 2010).

Climate change, as we have already noted, is only the most prominent among a host of predicted objects that offer new handholds for governance at supranational scales. Whether wrapped in the bureaucratic language of sustainable development goals or focused on outbreaks and manifestations such as international terrorism, the spectre of pandemics, or a 'Day Zero' of no water

in Chennai or Capetown, the future of human societies on Earth seems bound up with scenarios and outcomes that resist management according to the old rules. The future calls for new instruments of control, commensurate with its decentred, immaterial, and shape-shifting concerns. How has that call been answered?

Intervention on a planetary scale

To contain a problem, one can tackle it at either end, root out the causes or contain the effects. The politics of the future encompasses both forms of action, but techniques diverge depending on whether the focal problem is seen principally as one of human behaviour or one that primarily implicates physical and biological systems. Correspondingly, the emerging instruments of governance can be roughly divided between those that seek to modify how people act and those that seek to control the impacts of harmful collective action. The former have more to do with our imagination of origins, the latter more with consequences. Either way, the rising preoccupation with futures has dragged the policy world toward a new age of engineered solutions, though opinions differ as to whether the objects most in need of engineering are human actors and their behaviours or the biogeochemical systems within which the species so abundantly, and consequentially, proliferates.

Social theorists of the twentieth century were fearful of technology displacing human agency and crippling our capacity to act – a concern memorably foreshadowed in Max Weber’s vision of ‘machine production’ that determines the lives of those born into the ‘tremendous cosmos of the modern economic order’ and imprisons them in an ‘iron cage’ (Weber 1930[1905]). The worries of our era seem fixated more on the imperfect human drivers of the economic order. The concern is that humans, left to their own devices, are cognitively flawed and will act as agents of *un-reason*, whether through ignorance or, more dangerously, through biases built into the human brain (Tversky and Kahneman 1974; Kahneman 2011). This ‘finding’ of collective irrationality has given the impetus to engineering solutions that attempt to fix the future by fixing ourselves as agents of its creation. Its most powerful technique is the nudge (Thaler and Sunstein 2008), a set of devices with which the presumed rational ruler can impel people to override their biased minds and make defensible choices. The policy-maker’s job, then, is to make sure that people frame and make the right decisions in situations where their unguided instincts might lead them astray. Policy, especially economic policy, becomes in part an architectural project whose function is to design environments that promote reasoned choice-making.

Philosophical discussions of nudging have tended to focus on whether such top-down steering of human subjects preserves essential values of autonomy and dignity (Waldron 2014; Johnson 2016; Sunstein 2016), but these analyses leave unexplored the fluidity of intersubjective knowledge and its role in shaping the very terms of moral reckoning. What, after all, does autonomy mean in a world

that, since Weber's time, has bound people to machines in ever more intricate networks of dependence, and now threatens to blur the distinction between human and machine intelligence? And what, in any case, gives ruling elites the authority to know what is right or rational for the masses? The felt paternalism of nudging – a factor that its advocates mitigate by calling it *libertarian* paternalism – takes on further overtones of hubris when politics becomes predictive. For in the world of prediction, the outputs that guarantee a form of legitimacy to the politics of the present are no longer discernible. Gains and losses are removed to the sphere of promises, and their validity may rest more on ideology than visible evidence (Beckert 2019). Like sustainability or a stable climate, the fruits of predictive policy are neither accessible to those living in the present nor even attributable with any certainty to particular courses of action or inaction. The legitimacy of such promises rarely flows from the statements of experts alone, statements that in any case are loaded with the values of their time and place. Rather, legitimacy depends crucially on whether the promises appeal to collective imaginations that are already embedded in STIs circulating in society, such as the myth of the conquerable frontier in US politics or, more recently, the simple promise to 'make America great again'.

At first glance, attempts to make the future tractable to policy through technology seems to avoid the pitfalls of reengineering humans according to some putatively ahistorical and translocal model of rational behaviour. When mechanical tools are enrolled into risk avoidance or harm mitigation, the solution seems geared toward producing the kinds of outputs that people can easily evaluate and whose effects can therefore be judged in accordance with socially accepted criteria of goodness. Yet, in the predictive world, the efficacy of the technological fix bumps up against the same constraints of imperfect vision and ideological predisposition that render nudging so deeply suspect. Thus, geoengineering through solar radiation management, the most ambitious of imagined remedies for climate change (Keith 2013), derives from Cold War superpower imaginaries of cloud seeding and weather modification. It promises to keep today's leaders in technology and capital, and their epistemic and moral biases (Rayner 2013), in the driver's seat when assessing the nature and gravity of harm. Needless to add, technological fixes conceived on such grand scales are premised on tacit understandings of who has the right to frame and solve problems. They do little or nothing to redress the histories of inequality and injustice that gave rise to many global social ills in the first place.

Cutting across instrumental visions of how to manage the future, whether by manipulating human behaviour or by mitigating foreseeable harm, is the perennial question for democracy: with whose assent? Political philosophy has wrestled with the problems of representing future generations and groups that are not represented by nation states or other well-defined political unions (Thompson 2010). The added dimension that STS brings to the picture is the co-productionist insight that modes of intervention and modes of self-awareness are interdependent. Subjective senses of belonging to a polity are conditioned, in other words, by the very modes of representing and intervening

that presuppose a preexisting community. Predictive politics thus has to continually interrogate how the devices used in governance interact with modes of identity-making, lest seemingly neutral, objective instruments such as nudging fashion and legitimate the worlds they presume merely to discover and regulate. The tie-ins between instruments of governance and the constitution of political subjects forms the final section of this chapter.

Predictive politics and governable subjects

Regulatory devices, whether social or material, are designed with tacit, if culturally grounded, understandings of how human beings reason and behave. Otherwise, such devices could not easily be implemented or achieve the desired results. Built into any regulatory system, therefore, is a model of the reasoning individual and a theory of collective action, as well as technical ideas of how things function in the material world. So, nudge theory rests on psychological claims about how the human mind works when confronted with choices, as well as on social agreements that ‘problems’ such as inadequate saving or excessive eating should be alleviated. Similarly, advocacy for solar geoengineering assumes that natural cooling observed from volcanic eruptions can be replicated with artificial injection of aerosols into the Earth’s atmosphere. It also presumes that, confronted with the dire consequences of climate change, people will favour engineering the atmosphere as the lesser of two evils.

These ways of imagining natural and social expectations together illustrate the dynamics of co-production in future-making, but they do more. Regulatory regimes in effect impose on social life what Michel Foucault termed ‘the order of things’ – an epistemic and normative framework of control in which human subjects behave as the world seems to demand of them (Foucault 1970[1966]). In this sense, regulatory policy is a socio-technical imaginary that projects onto society desired futures toward which people are expected to direct their actions. Thus, a speed limit presupposes that people will not drive much faster than the sign says, a tax code takes for granted that most people will pay their financial dues, an environmental standard defines what people will regard as a safe industrial practice, and an anti-terrorism law expects that some, at least, will be deterred from extreme violence, knowing that massive penalties await them. Collective reason and even felt agency and subjectivity, in short, do not preexist the imagined order of the future. They are constituted and performed in the implementation of the imaginary.

For social scientists steeped in liberal thought the market has served as one powerful regulatory imaginary, indeed since the later twentieth century as the dominant one. It is the cornerstone of the architecture of choice. The (perfect) market is conceived as the ultimate unconstrained space in which people freely barter and make exchanges, ensuring that maximum collective well-being is secured at least cost and least constraint. Incentivizing rational, competitive behaviour by private entities is seen as the antidote to state failure. Markets, it is believed, unleash human energy in the search for better solutions, instead of

tamping down initiative through ill-conceived overreaching by poorly informed and resourced states. The alleged defeat of socialism in the late twentieth century offered ultimate vindication of these beliefs. Not surprisingly, then, market incentives have been proposed for managing some of the world's most intransigent future problems, carbon markets for controlling greenhouse gases being only the most prominent.

If, however, we see the market as a socio-technical imaginary rather than as a tool for regulating collective behaviour, the work needed to make markets function becomes more analytically salient. It is well known from the STS literature that the very idea of establishing an exchange system requires acts of commensuration that raise all the problems of judgement and reduction associated with government by numbers. To make markets work, one must establish which things can be exchanged for which other things. This is far from straightforward even when trades involve basic commodities such as clothing or food stuffs, but the difficulties mount exponentially when previously non-tradeable, physically disparate entities such as greenhouse gases are turned into goods to be exchanged (MacKenzie 2009). Commodities, moreover, must be standardized and rendered transparent for counts and audits in order to enable trust in a market – again, no easy task (Power 1997).

More importantly, however, making markets blurs the distinction between economics and politics and between consumers and citizens. Establishing an exchange system by making tradeable commodities serves at one and the same time as an exercise in controlling the status of political subjects. To the extent that market-making involves the production of things of value, it also involves a reshaping of what people want, and by extension the will of the subjects holding those preferences. Thus, it has become a truism to say that millennials who grew up in the digital era have a different sense of privacy from their elders, because they think nothing of sharing deeply personal information and images with tens and hundreds of others (Turkle 1995). Many indeed give up their personal data cheerfully because they welcome targeting by private goods and service providers whom they might not have encountered but for the infinitely webbed connections of the internet. In the biological realm, the development of genetic engineering and genome editing not only gave rise to a socio-technical imaginary of life liberated from inherited disease, but it also changed our understanding of conditions that human beings may rightfully wish not to be born with – today genetic disease, but tomorrow possibly short stature, dark skin, lower IQ, or depressive tendencies. It also changed politics. Human perfectibility may have been an aspiration since humans learned to envision a future, but in the domain of prediction it becomes part of a promissory package that governments can offer to win votes and sway opinion.⁸

The logic of the market demands, as well, a redrawing of the boundary between private and public. Sometimes that line is explicitly moved by law or regulation, as when a government decides to privatize what was a state obligation, such as a railway or a water supply, or in reverse to assume control over things and activities that once were privately managed, from healthcare to

environmental pollution to behaviour in the workplace. Such moments redefine the rights and responsibilities of states toward their subjects, whether by strengthening or weakening them. They are, straightforwardly, constitutional moments, often designed to correct widely recognized past wrongs.

In the predictive world, however, subjects and obligations are often constituted by less overt means, by redefining the parameters of social concern and social cohesion through developments in science and technology. Economists may have decided that temporally distant risks are nothing to worry about and should be discounted for purposes of action in the present. But climate science, with its powerful push to take planetary sustainability on board as a matter of human concern, upends economics and positions humanity as a steward with the responsibility to care for distant outcomes. As is apparent in the climate case, new tensions may thereby be called forth: between a more expansionist form of politics that willingly engages with bigger temporal and spatial problem definitions and a narrower politics of reaction and retrenchment that prefers to ignore predictions and rely instead upon a more immediate, and local, weighting of distributive outcomes. It is the politics of Greta Thunberg and the child activists against Trump and the American fossil fuel extractors.

Conclusion

Que sera sera, or what will be will be. We know these sayings. They represent for many of us the sense that futures are not ours to see or control. Indeed, for much of human history, the actual represented the limits of the possible. People could not dream of a very different future without invoking an afterlife, god and religion, sin and salvation, heaven and hell. If works on earth in the here and now could enable better future outcomes, those outcomes still were thought to lie in worlds that did not touch ours, across material bridges that were ours to build in the present. Politics, the struggle for worldly benefits, remained a matter of outcomes of the day, fought out in modern democracies within the enclaves of nation states, in accordance with rules of the game created by constitutional orders that knew the difference between public and private, consumer and citizen.

A series of developments in science and technology led from the nineteenth century onward toward a domestication of the future for present-day political action. Statistics allowed entire populations to be enumerated, mapped, and controlled, while computer simulations and the advent of big data created the illusion of a future where human needs and demands would ‘speak for themselves’ and all that politics would need to do would be to find the best solutions. Meantime, the nuclear age and the time bomb of climate change expanded the sphere of necessary political action from the national and international to the global, with associated demands for institutional and political innovation. A drift began toward a conception of politics as a matter of engineering – not normative struggle over the ends and purposes of life – with the state’s engineering instincts hovering between altering the political subject to fit the

rulers' allegedly rational ambitions and altering earth systems, including human biology, to produce distant, but universal, outcomes, such as sustainable development goals. At the same time, largely unregulated social media positioned the digital giants of the twenty-first century as data oligarchs, pitting the power of Google and Facebook against that of national governments in their capacity to constitute new objects of concern, new regulatory instruments, and new political subjects and axes of mobilization.

Predictive politics is here to stay, but the dynamics of the move from the politics of the world-as-it-is to that of worlds-yet-to-come demand a sharper awareness of the moves being made in this transition. Those moves, as argued in this chapter, include shifts in the definition of objects of governance, the development of instruments of intervention, and the constitution of political subjects. In this threefold process, older lines of political agency and self-awareness – most notably the line between public and private – are at risk of being obliterated without adequate critique or response. We need a better mapping of the terrain of predictive politics, and a clearer acknowledgement that the sites and objects of politics are no longer simply those of formal state institutions inherited from the early days of liberal democracy. Too much power to constitute the political now lies in the hands of institutions, deploying new forms of technocapital, that fall outside the bounds of any meaningful political control. Recognizing the domain of socio-technical imaginaries as a field of political action, and demanding the right forms of participation to shape those imaginaries, are essential steps toward reclaiming for this century's citizens an authentic politics of the future.

Notes

- 1 A well-known example from US politics is the failure of President George H. W. Bush to abide by his much-quoted 1988 campaign promise, 'Read my lips: no new taxes.' He was attacked from both right and left for his failure to hold the line on taxes and lost his bid for reelection in 1992 to the moderate Democratic candidate Bill Clinton.
- 2 These dates are admittedly somewhat artificial, but they correspond, respectively, to the start of the American Revolution, the Civil War, and the adoption of the Nineteenth Amendment (women's suffrage) to the US Constitution; the French Revolution; the Salt March led by Gandhi in India to protest Britain's oppressive Salt Act; and the anti-extradition protests in Hong Kong.
- 3 It is worth noting in this context that festering economic and social inequalities tied to race in the United States brought voting rights back to the top of the political agenda after 2016. The decision by failed gubernatorial candidate Stacey Abrams of Georgia to fight against voter suppression instead of herself running for election in 2020 illustrates the point being made here (Cobb 2019).
- 4 Winner argued famously, if controversially, that New York's famed planner Robert Moses did just this when he planned the highway overpasses leading to the local beaches too low to accommodate buses, the transport means of the poor, but to allow passenger cars, the means favoured by the city's rising middle and upper classes.
- 5 In the second edition of his extraordinarily influential book, Anderson identified the census, the map, and the museum as additional instruments for creating national

identity, complementing newspapers, the agents of 'print capitalism' that cause people of a given polity to read and identify with the same stories at the same time (Anderson 2006[1983]: 167–90).

- 6 In Anglo-American common law, courts were historically reluctant to hold private actors responsible for harms that were not 'imminent', which meant in practice legible to courts. These attitudes began changing toward the end of the twentieth century.
- 7 Extract from a speech by Swedish child activist Greta Thunberg at the UN General Assembly Climate Action Summit on 23 September 2019. For a full transcript, see www.npr.org/2019/09/23/763452863/transcript-greta-thunbergs-speech-at-the-u-n-climate-action-summit (accessed September 2019).
- 8 President Bill Clinton pointed the way toward this possibility during his remarks on the release of the first map of the human genome in June 2000:

Just to offer one example, patients with some forms of leukemia and breast cancer already are being treated in clinical trials with sophisticated new drugs that precisely target the faulty genes and cancer cells, with little or no risk to healthy cells. In fact, it is now conceivable that our children's children will know the term cancer only as a constellation of stars.

(White House, Office of the Press Secretary, 26 June 2000, <https://clintonwhitehouse3.archives.gov/WH/New/html/genome-20000626.html> (accessed September 2019))

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